EC417 – ADULT INCOME FAIRNESS AUDIT

GitHub Repository README

Project Overview

This repository contains a machine-learning project that predicts income categories using the Adult Income dataset and evaluates differences in prediction outcomes across demographic groups.

Objectives

• Data preprocessing and exploratory analysis
• Logistic Regression and Random Forest baselines
• CatBoost training and hyperparameter tuning
• Classification threshold optimization
• Fairness audit by sex and race
• Comparison with and without sensitive attributes
• Official test evaluation, error analysis and bootstrap confidence intervals

Dataset

Use the UCI Adult Income dataset. The notebook expects adult.data and adult.test. These raw dataset files are not included in this documentation PDF; obtain them separately and place them where the notebook expects them.

Repository Structure

```
EC417_Adult_Income_Fairness_Audit/
|-- README.md
|-- requirements.txt
|-- .gitignore
|-- EC417_Adult_Income_Fairness_Audit.ipynb
|-- src/
|   |-- __init__.py
|   |-- fairness_metrics.py
|-- report/
|   |-- project_report.md
|-- results/
    |-- test_model_comparison.csv
    |-- fairness audit CSV files
    |-- bootstrap confidence intervals
    |-- test error analysis
```

Installation

```
git clone <YOUR_GITHUB_REPOSITORY_URL>
cd EC417_Adult_Income_Fairness_Audit
pip install -r requirements.txt
jupyter notebook EC417_Adult_Income_Fairness_Audit.ipynb
```

Dependencies

numpy, pandas, matplotlib, seaborn, scikit-learn, catboost, jupyter, ipykernel, nbformat and tabulate.

Reproducibility Note

The uploaded notebook begins with a Google Colab file-upload cell. For local execution, replace that upload step with code that reads adult.data and adult.test from the local project directory. Run all cells in order.

Project Results

The saved notebook outputs report CatBoost ROC-AUC of approximately 0.927 on the official test set. Refer to the project report and generated CSV result files for the full model and fairness comparisons.